Sequences and Series

Section 1: Limits of Sequences

1 Find the limit of the sequence $a_n = \frac{3n^2 - 5n}{2n^2 + 7}$ as $n \to \infty$.

a. $3/2$

b. 0

c. ∞

d. $3/7$

2 Determine if the sequence $a_n = \frac{(-1)^n}{n}$ converges, and if so, find its limit.

a. Converges to 0

b. Diverges (oscillates)

c. Converges to 1

d. Converges to -1

3 Find the limit of the sequence $a_n = \left(1 + \frac{2}{n}\right)^n$ as $n \to \infty$.

a. e^2

b. e

c. 1

d. ∞

Section 2: Geometric Series

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4 Find the sum of the geometric series $\sum_{n=1}^{\infty} 3\left(\frac{2}{5}\right)^{n-1}$.

a. 5

b. 15/2

c. 3

d. 15

5 Determine the convergence of $\sum_{n=0}^{\infty} 4\left(-\frac{3}{2}\right)^{n}$.

a. Diverges because $|r| = 3/2 \geq 1$

b. Converges to $-8/5$

c. Converges to $8/5$

d. Converges to 0

6 Express the repeating decimal $0.\overline{27} = 0.272727\ldots$ as a simplified fraction using a geometric series.

a. 3/11

b. 27/100

c. 9/33

d. 27/99 is 3/11

Section 3: Telescoping Series

7 Find the sum of the telescoping series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

a. 1

b. 1/2

c. 2

d. Diverges

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8 Find the sum of $\sum_{n=1}^{\infty}\left(\frac{1}{\sqrt{n}}-\frac{1}{\sqrt{n+1}}\right)$.

a. 1

b. 0

c. ∞

d. $\sqrt{2}$

9 Evaluate $\sum_{n=1}^{\infty}\ln\left(\frac{n+1}{n}\right)$.

a. Diverges to ∞

b. 0

c. 1

d. $\ln 2$

Section 4: Divergence Test (nth-Term Test)

10 What does the nth-Term Test for Divergence state?

a. If $\lim_{n\to\infty} a_n \neq 0$, then $\sum a_n$ diverges

b. If $\lim_{n\to\infty} a_n = 0$, then $\sum a_n$ converges

c. If $\sum a_n$ diverges, then $\lim_{n\to\infty} a_n \neq 0$

d. If $\lim_{n\to\infty} a_n = 0$, the test proves divergence

11 Apply the Divergence Test to $\sum_{n=1}^{\infty}\frac{n}{2n+1}$.

a. Diverges because $\lim_{n\to\infty} a_n = 1/2 \neq 0$

b. Converges to $1/2$

c. Inconclusive

d. Converges by p-series

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12 Why is the Divergence Test inconclusive for the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$?

a. Because $\lim_{n \to \infty} \frac{1}{n} = 0$, which does not guarantee convergence

b. Because $1/n$ is negative

c. Because the terms oscillate

d. Because it is a geometric series

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